**Q1.** (a) Debbie put a paper cup into a glass beaker.
She glued a magnet in the bottom of the paper cup.
She glued another magnet in the bottom of the beaker.
The magnets repelled.

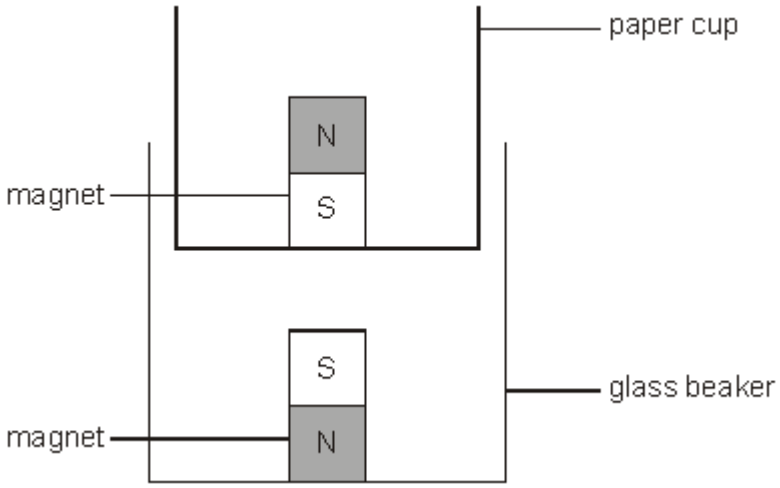



**diagram A**

*not to scale*

What **two** forces act on the paper cup and its contents to keep it in this position?

1. .....

1 mark

2. .....

1 mark

(b) Debbie put 5 g of aluminium rivets into the paper cup.
It moved down a little as shown in diagram B.

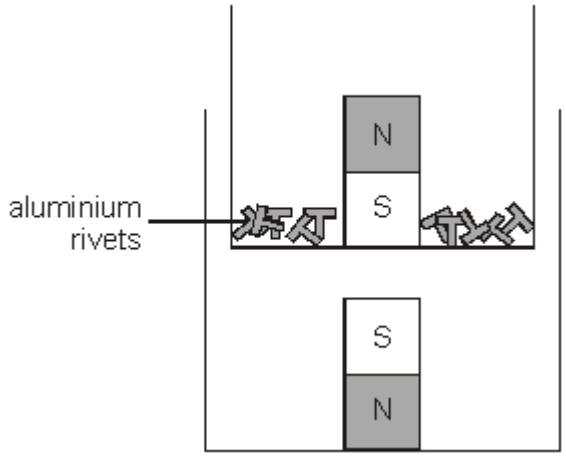
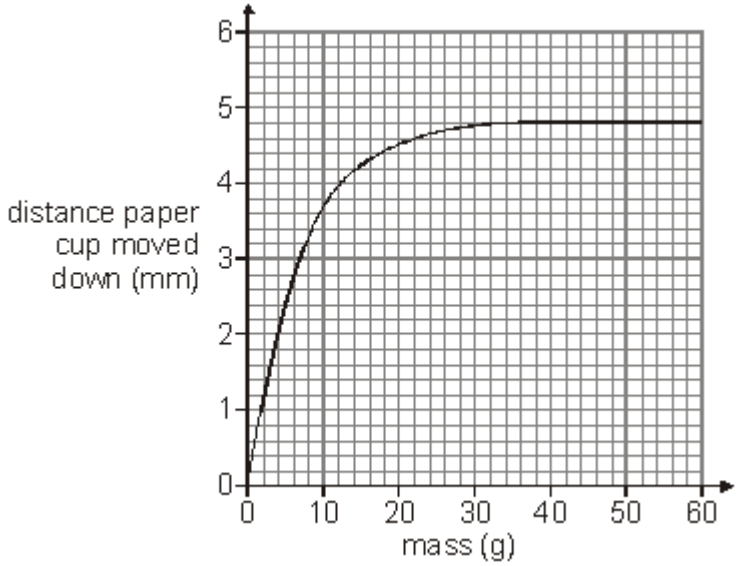



**diagram B**

*not to scale*

Debbie plotted a graph to show how the mass of aluminium rivets affected the distance the cup moved down.




(i) Use the graph to find the mass that made the cup move down 4 mm.

..... g

1 mark

(ii) Why did the graph stay flat with masses greater than 40 g?

.....

1 mark

(c) Debbie removed the 5 g of aluminium rivets and put 5 g of iron nails into the cup.

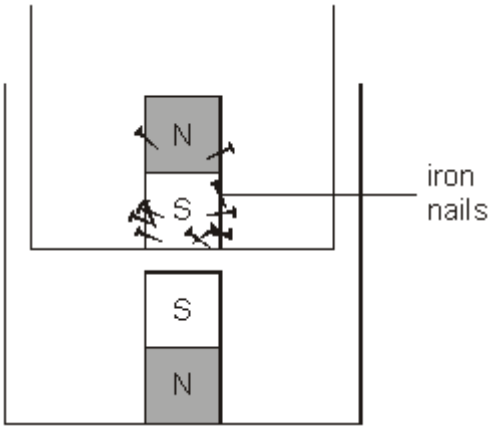



**diagram C**

*not to scale*

The paper cup moved down more with 5 g of iron nails than with 5 g of aluminium rivets as shown in diagram C.
Give the reason for this.

.....

.....

1 mark
maximum 5 marks

Mark schemes

## Q1.

(a) gravity

*accept 'weight'*

**1 (L5)**

magnetic force **or** magnetism

*accept 'repulsion' **or** 'upthrust'*
*answers may be in either order*
*do **not** accept 'air resistance'*

**1 (L5)**

(b) (i) 12

**1 (L5)**

(ii) any **one** from

- the paper cup stopped moving
  *accept 'it hit the bottom'*
- the paper cup reached the bottom magnet
  *accept 'the paper cup could not go any further'*

**1 (L6)**

(c) any **one** from

- iron is magnetic
  *accept 'aluminium is not magnetic'*
- iron nails are attracted to a magnet
  *accept 'the rivets are not attracted to a magnet'*
- there is a magnetic force on the iron
  *do **not** accept 'aluminium **or** rivets are less magnetic'*
  *do **not** accept 'iron **or** nails are more magnetic than aluminium*
  ***or** rivets'*

**1 (L6)**

**[5]**